# The length of an SS Cygni outburst helps forecast the next: a preregistered test on a century of amateur observations

Barry W. Brook*[1]

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1. University of Tasmania, Hobart, Tasmania, Australia

**Corresponding author:** Barry W. Brook

## Abstract

Dwarf novae brighten in repeated outbursts when the gas disc around a white dwarf heats up. In SS Cygni, longer outbursts tend to precede longer quiet intervals. Using a century of amateur observations from the AAVSO, we tested whether outburst duration improves forecasts of the next onset beyond recent timing. Lognormal recurrence models fitted on 1910–1959 were frozen and preregistered before 1960–2017 was opened. Duration raised the working log score by 0.0165 nats per forecast (90% interval 0.0049–0.0281; 441 forecasts), meeting the preregistered criterion. Two redesigned timings and both preregistered secondary tests gave smaller gains with intervals spanning zero. A frozen screen of ten other dwarf novae found none whose pre-1990 records could repeat the test. Linking the working score to physical onsets requires a validated observation model, and shared end estimates or activity-dependent observing could couple duration with timing. A documented parser repair followed first data access.

**Key words:** stars: dwarf novae – stars: individual: SS Cygni – accretion, accretion discs – methods: statistical

## 1 Introduction

Dwarf novae are close binaries in which a white dwarf accretes gas from a low-mass companion through a disc. Every few weeks to months the disc switches from a cool, faint state to a hot, bright one, and the system brightens by several magnitudes for days to weeks before fading. The disc instability model attributes these outbursts to a thermal–viscous instability driven by hydrogen ionisation (Lasota 2001; Hameury 2020). In that picture an outburst drains part of the disc onto the white dwarf, and the next outburst begins once the disc has refilled to a critical state.

SS Cygni (SS Cyg) is one of the brightest and best-observed dwarf novae. Amateur observers have reported its brightness to the American Association of Variable Star Observers (AAVSO) since the 1890s, producing a nearly continuous visual light curve. Its outbursts recur roughly every seven weeks and fall into short and long kinds, lasting about one and two weeks respectively (Fig. 1A).

The record has long been searched for dependence between successive cycles. Bath and van Paradijs related outburst energy to the following interval (Bath and van Paradijs 1983). Cannizzo and Mattei (CM92) analysed 705 outbursts and found that duration correlates with the following quiet interval ($r = 0.223$) and with the preceding one ($r = 0.282$), that successive durations alternate ($r = -0.400$), and that durations divide at about 12 days into short and long classes (Cannizzo and Mattei 1992). They also showed that how a cycle is defined governs the strength of duration–cycle correlations, and later extended the outburst statistics (Cannizzo and Mattei 1998). Disc calculations reproduce long/short alternation and slow changes in recurrence (Cannizzo 1993). Related work examined symbolic dependence in outburst widths (Hempelmann and Kurths, *Astronomy and Astrophysics*, 1990, 232, 356), searched for low-dimensional chaos (Cannizzo and Goodings 1988) and short-term precursors (Price et al. 2007), and studied recurrence in other dwarf novae (van der Woerd and van Paradijs 1987).

These studies estimate associations on the data they describe. A forecast test asks whether the information improves predictions of data the forecaster has not seen. The answer depends on what the forecaster already knows: recent quiet intervals carry information about the next one, so the useful question is whether duration adds to them. The closest forecasting precedent we found is Grigorievskiy and colleagues' study in the 2014 ITISE proceedings (pp. 520–531), which tested point forecasts of onset on one held-out outburst of V363 Lyr. To our knowledge, the present study is the first preregistered probabilistic test of dwarf-nova recurrence forecasts on held-out data, and the first held-out test of duration beyond a timing baseline.

Two hypotheses predict that duration should help. Under the *depletion hypothesis*, a long outburst drains more of the disc, which then takes longer to refill when the gas supply from the companion is similar from cycle to cycle. Under the *state hypothesis*, duration tracks a slowly changing accretion state, such as the mass-transfer rate, that recent intervals describe incompletely. Both predict that forecasts including duration will outperform forecasts based on timing alone. Separating them requires measurements beyond outburst timing (Section 4.2).

We tested the prediction with a design that is standard in forecasting and rare in astronomy. Forecasts

were judged by a logarithmic score, which rewards the probability assigned to what happens (Gneiting and Raftery 2007), and evaluated strictly forward in time (Bürkner, Gabry, and Vehtari 2020). We fitted the models to 1910–1959, froze them, published the protocol and then scored every forecast for 1960–2017. We then examined how the result depends on the way outbursts are timed, and screened ten other dwarf novae to see whether their records could support the same test. Section 2 describes the data and methods, Section 3 the results, Section 4 their interpretation and Section 5 the limitations.

## 2 Data and methods

### 2.1 Observations

We used visual magnitude estimates of SS Cyg from the AAVSO International Database, requested in bounded annual windows, hashed on receipt and restricted to the visual band (`Vis.`) with validation flag `V`. Reports fainter than an observer's detection limit were kept as limits. All model choices and fits used the development period 1910–1959. The evaluation period covers forecasts issued from 1960 January 1 to 2017 December 31 (JD 2436934.5 to 2458119.5); it contains 500,479 archive rows after deduplication, of which 249,356 meet the selector. Observations from 2018 onward are reserved for a future test. Times are Julian Dates as returned by the archive.

### 2.2 Outburst extraction

An extraction algorithm, X4, identifies outbursts as episodes brighter than visual magnitude 10, which lies between SS Cyg's quiescent and outburst brightness. X4 groups each observer's reports into visits and estimates when the light curve crossed magnitude 10 on the rise (the onset) and on the fade (the end). Each crossing is summarised as an interval and a point, fitted to nearby reports with a shared 0.05-mag allowance for systematic error common to observers. An episode is confirmed by bright reports from two distinct observers, or from one observer on at least two nights, with a margin past the threshold and no contradicting faint report. Duration runs from fitted onset to fitted end, and a quiet interval from one outburst's end to the next onset. A quiet interval counts as covered when no gap between visits within it exceeds seven days. X4 was adopted during development after three earlier extractors failed a synthetic recovery screen; it passed that screen at magnitude 10 (Appendix A).

### 2.3 Forecasts

A forecast is issued at the observation time when reports first confirm that the current outburst has faded. Every such confirmation issues a forecast, including repeats. Each forecast uses only reports dated within the preceding 400 days: X4 is rerun on that window alone, and tests that insert later rows confirm that no predictor changes.

Each forecast is a lognormal distribution for the quiet interval $Q$ that follows the fitted end $e$ of the completed outburst. With log-location $\mu$ and log-scale $\sigma$, the three models are

$$\begin{aligned} \text{M0}: &\ \mu = \alpha, \\ \text{M2}: &\ \mu = \alpha + \gamma_1 q + \gamma_2 L, \\ \text{M3}: &\ \mu = \alpha + \gamma_1 q + \gamma_2 L + \beta x, \quad \beta \geq 0, \end{aligned}$$

where $q$ is the standardised log of the previous covered quiet interval, $L$ the standardised mean of up to six earlier covered log quiet intervals ($L = 0$ when none is available), and $x$ the standardised log duration. Centres and scales are development means and standard deviations. M2 is the timing baseline and M3 adds duration; the constraint $\beta \geq 0$ encodes the direction both hypotheses predict. A forecast issued at age $a = t_{\text{issue}} - e$ is the lognormal conditioned on $Q > a$. Forecasts without a covered previous interval use M0 in both arms, and forecasts without a fitted end use a common remaining-wait forecast; both contribute exactly zero to the comparison. Table 1 summarises the models and test, and Fig. 1B shows one cycle schematically.

### 2.4 Outcomes and the working score

The outcome of a forecast is the onset of the next outburst as X4 estimates it from the later record, an interval $(l, u]$. A forecast stops at the first loss of coverage (a visit gap over seven days), at the next issued forecast, or at the end of the evaluation period. If no onset is established by then, the forecast is right-censored at that time $c$. The *working score* of a forecast is

$$s = \begin{cases} \log\left[S(l-e) - S(u-e)\right] - \log S(a) & \text{onset interval,} \\ \log S(c-e) - \log S(a) & \text{censored,} \end{cases}$$

where $S$ is the lognormal survival function. It is the log probability the forecast assigned to the extracted onset window, or to no onset before $c$, measured in nats; higher is better. The paired gain is M3's score minus M2's for the same forecast. Exponentiating a mean gain gives the ratio of the geometric-mean probabilities the two models assigned to the scored outcomes.

### 2.5 Fitting, freezing and the test

M2 was fitted by maximum likelihood to the 307 timing-eligible development forecasts and frozen. The duration coefficient was then fitted with M2 held fixed, so that $\beta = 0$ reproduces M2 exactly. The frozen values are $\beta = 0.0466$ per standard deviation of log duration, $\gamma_1 = 0.0780$, $\gamma_2 = 0.0513$ and $\sigma = 0.346$.

The single preregistered statistic is the mean paired gain over all score-eligible forecasts issued in 1960–2017, including the zero contributions of fallback forecasts. Its standard error is cluster-robust over consecutive six-year blocks (2191.5 days) from the start of 1960, with a $G/(G-1)$ correction for $G$ nonempty blocks, and inference uses Student's t with $G-1$ degrees of freedom. The criterion was a one-sided 95% lower bound above zero, equivalent to a two-sided 90% interval wholly above zero. The protocol also specified secondary comparisons: an era split at 1993, after the end of CM92's record, and a rule that replaces continuous duration with CM92's 12-day long/short classes plus within-class duration. Before the development fit, a synthetic screen confirmed 11 of 40 simulated evaluations at an effect sized to CM92's correlations, and 3 of 220 no-effect histories.

### 2.6 Access and execution

The protocol was deposited on Zenodo (doi:10.5281/zenodo.24053338) on 2026 November 8, before any evaluation data were requested (Fig. 1C). The first request stopped before any forecast was scored, because the archive returned the literal token `None` for missing uncertainties, a field that no analysis uses. The parser was changed to accept that exact token without altering or dropping a row. Independent review, invariance tests and a full synthetic rehearsal passed before the run resumed, and all forecasting code, parameters and protocol files stayed unchanged. Two independent implementations of the forecast ledger had to agree exactly before scoring, and a separate checker reproduced every score.

### 2.7 Measurement checks after the result

After the result, an external review asked how far it depends on the way outbursts are timed. The analyses in this subsection were each specified before their own fits, and all reuse the opened data.

- **Rescoring (R1)** audited the censoring rules, scored each forecast against X4's point estimate of onset in place of its interval, and added synthetic errors to fitted end times.
- **Report windows and clocks (R2b)** rebuilt each onset outcome as a report window, from the last faint report to the first decisively bright one (extractor X5). Two clocks were compared. The first keeps the fitted end. The second starts the quiet interval at the first decisively faint report and measures duration as the span of decisively bright reports, so no end time is interpolated. These comparisons use new windowed timing baselines: B0 with recent-interval covariates, B1 with discounted history and B2, a state-space level model chosen in advance.
- **Drift recalibration** recentred duration on the preceding ten outbursts and updated the baseline level from preceding outcomes, to ask how changes in the typical quiet interval affect the gain.

### 2.8 Transfer screen

To ask whether other dwarf novae could support the same test, we froze a screen in Git before requesting any of their data. Eight candidates (SS Aur, RU Peg, AR And, X Leo, TW Vir, UU Aql, CH UMa and DX And) and two eclipsing systems held to stricter rules (U Gem and IP Peg) were chosen from catalogue properties, with thresholds proposed from the same catalogues. Only records before 1990 (JD $<$ 2447892.5) were opened, keeping later decades for a future test. A star qualified if its pre-1990 record yielded at least 40 supported forecasts (with a covered previous interval, a measurable duration and a scorable outcome), at least 70% eligible onset windows, stable timing-baseline fits, and no bright run of 60 days or more; the eclipsing systems also needed to pass an eclipse-handling check. The screen fitted no duration effect and computed no forecast gain (Appendix E).

**Table 1.** Models and test.

| Item | Specification |
|---|---|
| Forecast | Lognormal quiet interval after the fitted end, conditioned on survival to issue |
| M0 | Constant log-location |
| M2, timing baseline | Log previous covered interval; mean of up to six earlier covered log intervals |
| M3 | M2 plus log duration, $\beta \geq 0$, fitted with M2 fixed |
| Development fit | 1910–1959; frozen $\beta = 0.0466$, $\gamma_1 = 0.0780$, $\gamma_2 = 0.0513$, $\sigma = 0.346$ |
| Evaluation | Every forecast issued 1960–2017 |
| Working score | Log probability of the extracted onset window, or of survival to censoring, given survival to issue |
| Statistic and criterion | Mean M3 − M2 gain; six-year cluster standard error; one-sided 95% lower bound above zero |
| Redesigned comparisons | Report-window outcomes; fitted-end and decisive-faint clocks; baselines B0, B1, B2 |

## 3 Results

### 3.1 Duration improved the frozen forecasts

The duration model forecast longer waits after longer outbursts: doubling the duration, roughly the difference between a short and a long outburst, lengthens the median of the lognormal clock by about 9%. Over 441 forecasts issued in 1960–2017, adding duration raised the mean working score by 0.0165 nats (90% interval 0.0049 to 0.0281; ten blocks, nine degrees of freedom), meeting the preregistered criterion (Fig. 2). The geometric-mean probability assigned to the scored outcomes rose by 1.7%. The timing baseline M2 itself gained 0.117 nats over the constant forecast M0, so duration added about 14% of what recent timing adds. The outcomes comprised 435 onset intervals and six censored forecasts, and five forecasts used the shared fallback.

Fig. 2A shows the first eligible forecast of the evaluation period, chosen by date. M3 gave a median remaining wait of 26.77 days against 27.57 for M2, and placed 0.603 of its probability within 30 days against 0.577. The gain was positive in eight of ten six-year blocks (Fig. 2B), and leaving out any one block gave means from 0.0132 to 0.0194. Frozen sensitivities to uniform end times, a previous-gap-only baseline, a two-day issue delay and three-year clusters kept the lower bound above zero (Appendix B). M3's predictive intervals were conservative: its nominal 50% intervals covered between 0.546 and 0.617 of outcomes, and its nominal 90% intervals between 0.923 and 0.980.

The preregistered era split gave 0.0240 nats (0.0091 to 0.0389; 261 forecasts) for 1960–1992 and 0.0057 (−0.0107 to 0.0221; 180 forecasts) for 1993–2017. The long/short rule gained 0.0025 nats (−0.0022 to 0.0072).

### 3.2 Measurement checks

Fig. 3 collects the duration gains from every comparison. Scoring against X4's point estimates of onset gave 0.0154 nats (0.0034 to 0.0274). The censoring audit changed zero of the 441 outcomes. The report-window comparisons gave smaller gains with intervals spanning zero: 0.0075 nats (−0.0008 to 0.0159; 432 forecasts) on the fitted-end clock and 0.0043 nats (−0.0008 to 0.0094; 431 forecasts) on the decisive-faint clock, both with baseline B2. Their paired difference was −0.0031 nats (−0.0078 to 0.0017; Appendix C).

In synthetic null and gap-dependent histories, adding independent errors to fitted end times gave mean gains of zero or below at every tested error size. An observing-rate covariate, the reporting rate during the completed outburst, gained 0.0030 nats (0.0013 to 0.0048) on the decisive-faint clock; adding it alongside duration left the duration gain at 0.0043 nats.

Of the redesigned baselines, B1 fitted the development data best and B2 worst (negative log likelihoods 827.24 for B1, 830.19 for B0 and 849.28 for B2). The typical quiet interval early in the evaluation period was shorter than in the development blocks. Recalibrating for that drift reduced the duration gain in the R2b comparison closest to the primary design (point clock, baseline B0) from 0.0184 to 0.0132 nats and the observing-rate gain from 0.0030 to 0.0014. After recalibration the duration gains stayed positive on every clock and baseline, from 0.0052 to 0.0132 nats; the point-clock B0 gain kept a positive lower bound.

### 3.3 Transfer screen

None of the ten stars qualified (Fig. 4; Appendix E). Nine yielded fewer than the 40 supported forecasts required, and across all ten the frozen pre-1990 streams yielded 217. SS Aur yielded 110, with 79.6% eligible onset windows, and failed the long-run check with a 100.97-day bright run made up of four separate clusters of report nights. The eclipsing systems U Gem and IP Peg yielded 27 and 0 supported forecasts, and their eclipse check stopped at its baseline stage.

The candidates' records time successive outbursts far less often than SS Cyg's. In 1980–1989 SS Cyg's record yielded 67 supported forecasts against 73 confirmed outbursts; the ten candidates together yielded 102 against 485 (Fig. 4B). The largest single loss was the requirement for a covered previous quiet interval: in an ordinary report-level diagnostic, 822 pre-1990 candidate forecasts with a timing clock fell to 280 once that requirement applied. At the 1980s rate, all ten stars together, ignoring the other qualification rules, would yield 367.2 supported forecasts over 36 years.

## 4 Discussion

### 4.1 What the result shows

The duration of a completed outburst improved forecasts of SS Cyg's next onset beyond what recent quiet intervals provided, in a test whose models and criterion were fixed before the evaluation data were opened. The direction matches CM92's correlation between duration and the following quiet interval, now carried forward in time over 58 years of observations. The gain is modest: about a seventh of what recent timing contributes. Most of the predictability of SS Cyg's next outburst lies in its recent rhythm, and duration adds a small increment, positive in eight of ten six-year blocks.

Rescoring against point onsets reproduces the gain. Rebuilding the outcomes from report windows gives smaller gains, 0.0075 and 0.0043 nats, with intervals spanning zero. The observing-rate covariate carries a smaller amount of forecast information of its own, and adding it leaves the duration gain unchanged on the decisive-faint clock.

### 4.2 Depletion or state?

Both hypotheses predict the observed direction. The depletion hypothesis ties the effect to individual cycles: each long outburst should lengthen the interval that follows. Disc models that drain different fractions of the disc in long and short outbursts reproduce the observed alternation (Cannizzo 1993), and alternation is strong in these data: the lag-one correlation of log duration was −0.52 in the evaluation years. In exploratory fits a long/short indicator described the evaluation data at least as well as continuous duration (Appendix D), which fits a cycle-level mechanism tied to outburst type.

The state hypothesis ties the effect to slower change: duration would carry information about the mass-transfer rate or disc state that the recent-interval average smooths over. SS Cyg's accretion state does change, as its anomalous event of 2021 showed (Kimura et al. 2021). The drift in the typical quiet interval between development and evaluation, and the smaller gains after recalibrating for it, fit a contribution from slow change, in the star or in how it was observed.

Forecasts that include an independent measure of accretion state, such as quiescent brightness or X-ray and extreme-ultraviolet flux (Wheatley, Mauche, and Mattei 2003), could separate the two: if duration's gain shrinks once state is known, the state hypothesis gains support; if it persists, depletion does. The frozen forecasts can also be scored on SS Cyg's observations from 2018 onward, which remain unopened.

### 4.3 What a record needs

The test was possible because SS Cyg's record times most successive outbursts. A forecast built on the previous interval needs uninterrupted coverage across two cycles, and one observing gap of more than a week removes it. Under the same rules the candidates' pre-1990 records yielded about one supported forecast per five confirmed outbursts, against nearly one per outburst for SS Cyg. Their records from 1990 onward, and automated surveys, may time outbursts more completely; a multi-star test would need its own frozen design on those records.

The design itself is general. Fitting on early records, freezing and publishing the forecasts, and scoring them on later records suits other long, irregular records of recurrent events kept by many observers, such as volcanic eruptions or disease outbreaks. It needs a record that times successive events, a baseline that captures their recent rhythm, and a score fixed before the test data are seen.

## 5 Limitations

Table 2 lists the assumptions behind the primary result, what tested each and what remains open. Three matter most.

First, the working score treats X4's onset windows as true. The windows are fitted statistical summaries, and linking the score to the physical onset of an outburst requires a validated model of how reports record the star: correct association of reports with outbursts, containment of the true crossing in each window, and an observation process independent of the outcome (Appendix C). X4 passed its recovery screen on waveforms known during development; recovery on a waveform family held out from development remains untested, because that benchmark failed qualification before X4 was scored on it.

Second, duration and the following quiet interval share one estimated end time, so errors in that time can couple them, and observers may report more often when the star is active. The decisive-faint clock removes interpolation from the end but keeps report timing, and an invented cadence construction shows that report timing alone can produce a positive duration gain. The endpoint-error experiments covered independent errors only, and the observing rate itself carried forecast information.

Third, everything after the primary result reuses opened data and is exploratory. The redesigned comparisons and the drift recalibration change baseline, clock, coding and scored population together, so their smaller gains measure no fraction of the primary gain. A richer timing baseline could also absorb part of that gain.

The evaluation followed a parser repair made after the first data access and before scoring; the protocol DOI timestamps the original code, and the repair is documented separately. Forecasts use reports by observation date, and when historical reports were uploaded or revised is unknown, so the test replays the archive as it stands today. The transfer screen's ratios count forecasts and outbursts by different landmarks and describe yield under its frozen rules, which used catalogue thresholds. The result concerns one star.

**Table 2.** Assumptions behind the primary result.

| Assumption | Why it matters | What tested it | What remains open |
|---|---|---|---|
| Extracted onset windows stand for physical onsets | Defines what the score rewards | Synthetic recovery screen; point rescoring; report windows | A validated observation model; recovery on unfamiliar waveforms |
| End-time errors are independent of the next interval | A shared end couples duration and gap | Sign constraint; independent-error experiments; decisive-faint clock | Systematic or state-dependent errors; report-timing coupling |
| Observing effort is independent of the outcome | Observers may watch an active star more closely | Observing-rate covariate | The rate itself carries information |
| The timing baseline is adequate | The gain is measured beyond it | Baselines B0–B2; drift recalibration | Richer baselines; recalibration is exploratory |
| The archive reflects what was reported at each date | Defines the information at issue | Prefix-only predictors; inserted-row tests | Upload and revision dates are unknown |
| Six-year blocks are independent | Sets the interval width | Leave-one-block-out; three-year clusters | A standard error 1.43 times larger removes the result |
| The evaluation followed the protocol | Confirmatory status | DOI before access; two implementations; independent checker | Parser repair after first access |
| One star | Scope | Ten-star transfer screen | No other star qualified |

## 6 Conclusions

A century of amateur observations of SS Cygni supported a preregistered test of a question first raised by correlations in the same record: does the length of an outburst help forecast the next? Forecasts including duration, frozen on 1910–1959 and scored on 1960–2017, assigned more probability to the extracted onsets than forecasts based on recent timing alone, meeting the preregistered criterion. The gain is a small increment on a rhythm that recent timing already captures well. Separating disc depletion from a slowly changing accretion state, and repeating the test on other dwarf novae, will need records that time successive outbursts as completely as SS Cyg's, together with independent measures of the accretion state. The frozen forecasts, code and protocol are public and ready to be scored on SS Cyg's later observations.

## Acknowledgments

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*[Placeholder: generative-AI declaration, to be written before submission. It will state the roles of GPT-6 Pro, Claude Opus 5.5 and GPT-6 Codex, and how their calculations, analyses and figures were tested and validated, as the journal requires.]*

## Data availability

The repository https://github.com/BWBrook/dwarf-nova-memory contains the protocols, source code, frozen parameters, aggregate evidence, claims ledger, figures and access receipts. Code is MIT licensed; text, figures and derived data are CC BY-SA 4.0. The evaluation protocol is archived at https://doi.org/10.5281/zenodo.24053338. AAVSO observations are available from the International Database. Raw observations, observer identities, event catalogues and individual forecast scores are retained outside the repository; reproduction uses those hashed inputs without a new archive request.

## Appendix A: Extraction, issuance and access

**Extraction.** X4 (`src/dnmem/extract_x4.py`) forms its crossing summaries from normalised slope-mixture fits to nearby estimates. The 0.05-mag shared allowance stops the intervals shrinking without limit as estimates accumulate; it was chosen on development simulations. The issuing rule arms only on bright evidence at least 0.5 mag past the threshold. Three earlier extractors, X1–X3, failed qualification. X4 passed a synthetic screen at magnitude 10 on the qualified cadence family, where the seven-day coverage rule was also chosen; magnitude-9.5 diagnostics failed in later eras.

**Issuance and predictors.** The current episode is the last prefix episode whose end bracket reaches the issuing bright evidence. A duration of zero or less makes the mark unavailable. The previous interval counts only if the previous episode begins more than 30 days inside the 400-day window. A two-day reporting delay was a frozen sensitivity.

**The sign constraint.** Shared end-time error pushed fitted null duration coefficients negative in development simulations, which motivated fitting $\beta$ with M2 fixed and bounding it at zero. The bound restricts direction; an exact construction with constant true duration can still yield positive information, so endpoint coupling remains a stated limit (Section 5).

**Access timeline** (2026 November 8, UTC). Protocol published 12:34:47; first request 12:38:28, stopped 12:39:51 before any ledger or score; repaired acquisition 12:51:18; ledger concordance and primary result 13:06:58. The repair freeze is commit `5e39aeb` and the execution commit `d7151e0`. No person or agent inspected real scores, outcomes, counts or coverage summaries before the primary result was written. A separate checker reproduced all 2,641 variant score rows to order $10^{-13}$.

## Appendix B: Frozen sensitivities and calibration

**Table B1.** Frozen comparisons: mean paired working-score gain (nats per forecast) and 90% cluster interval. Only the first row is confirmatory.

| Comparison | Forecasts | Gain | Interval |
|---|---|---|---|
| Primary, M3 − M2, 1960–2017 | 441 | 0.0165 | 0.0049 to 0.0281 |
| 1960–1992 | 261 | 0.0240 | 0.0091 to 0.0389 |
| 1993–2017 | 180 | 0.0057 | −0.0107 to 0.0221 |
| 12-day class and within-class rule | 441 | 0.0025 | −0.0022 to 0.0072 |
| Uniform-end marks | 441 | 0.0166 | lower 0.0044 |
| Previous-gap-only baseline | 441 | 0.0170 | lower 0.0054 |
| Two-day issue delay | 441 | 0.0165 | lower 0.0049 |
| Joint-valid forecasts | 436 | 0.0167 | lower 0.0049 |
| Three-year clusters | 441 | as primary | lower 0.0059 |
| Baseline M2 − M0 | 441 | 0.1168 | 0.0663 to 0.1672 |

Deleting the six censored forecasts gives 0.01678 nats. Coverage bounds for M3's central intervals (Fig. 5) run from definitely to possibly covered, given the extracted windows and censoring; they describe this sample. The frozen placebo with a lagged duration equals M2, because its development coefficient sat at the bound of zero.

## Appendix C: Report windows, clocks and baselines

**Report windows.** X5 searches for decisive faintness after the previous peak before selecting the next decisive-bright rise, and keeps raw time supports and flags. It yielded 818 available and 805 eligible windows among 845 confirmed episodes, with median width 1.15 days. The original report clips were missing for 434 of the original associations, so the earlier report-interval rescoring covered only 11 contrasts.

**Premises.** Treating a report window as the onset outcome rests on three premises: correct association of reports with outbursts, completeness and a valid risk set (A0); containment of the true crossing in the window, or genuine survival through a censoring time (A1); and a complete, normalised observation process that is coarsened at random (A2). Under them the report likelihood factors into an observation term and the forecast probability of the window. Visit timing and margins past threshold do not by themselves establish the premises.

**Clocks.** The decisive-faint clock anchors each forecast at the first decisively faint report, uses the decisive-bright span as duration, and waits for report availability before issuing. Exact coincident issues share one representative, and unsupported outcomes stay unsupported.

**Table C1.** Report-window contrasts. H3 is the paired difference between the H2 and matched H1 gains.

| Exploratory contrast | Scored issuances | Mean gain, nats | Six-year 90% interval | Twelve-year 90% interval |
|---|---|---|---|---|
| H1: fitted-end clock, B2 duration | 432 | 0.0075 | −0.0008 to 0.0159 | −0.0014 to 0.0165 |
| H2: observed clock, B2 bright span | 431 | 0.0043 | −0.0008 to 0.0094 | −0.0021 to 0.0107 |
| H3: observed-clock gain minus matched fitted-end gain | 431 | −0.0031 | −0.0078 to 0.0017 | −0.0078 to 0.0017 |

**Baselines and drift.** The drift recalibration subtracted the mean of the preceding ten marks and recalibrated the baseline level from preceding evaluation outcomes, keeping the development coefficient. It used final saved labels at proxy availability times, and it is a diagnostic of the historical record, distinct from a prospective forecasting algorithm.

## Appendix D: Long/short classes

With M2's historical parameters fixed and a free common level, likelihood-ratio statistics on the evaluation data were 14.7 for the class indicator against level only, 11.7 for continuous log duration against level only, and 0.17 for adding within-class duration to the class indicator. The class contrast corresponds to a 14% larger working clock scale after long outbursts (Fig. 6). It was 0.16323 (statistic 13.504) in 1960–1992 and 0.07828 (2.262) in 1993–2017. Giving both frozen rules M2's evaluation-fitted level raised the class rule's gain to 0.00905 nats (0.00409 to 0.01401). A long outburst followed a long one in 0.1495 of forecast-record pairs and a short one in 0.6640. These fits reuse the evaluation outcomes.

## Appendix E: Transfer screen

The screen's code was fixed in Git at commit `d827313` before any candidate record was requested. Names were resolved under the same date restriction, and pagination, identity, row-window and duplicate checks preceded analysis. Supported forecasts used the decisive-faint clock and needed a positive mark and an eligible onset window or positive-exposure censoring. Baseline stability was tested on B0 fits without marks. The long-run rule flags a supported bright run of at least 60 days, or repeated 40-day runs, with minimum night and observer support. The eclipsing systems used two derived observer-night streams, which had to agree, and a synthetic eclipse test whose no-eclipse baseline had to recover enough events first; neither system passed that baseline.

**Table E1.** Frozen qualification. Eclipsing-system counts use the median derived-night stream.

| Candidate | Supported development issuances | Eligible onset windows | Main failed conditions |
|---|---|---|---|
| SS Aur | 110 | 79.6% | Operational morphology |
| RU Peg | 23 | 63.5% | Count, windows |
| AR And | 7 | 31.4% | Count, windows, base information/fit |
| X Leo | 35 | 44.0% | Count, windows |
| TW Vir | 0 | 11.6% | Count, windows, base information/fit |
| UU Aql | 15 | 80.8% | Count |
| CH UMa | 0 | 70.4% | Count, base information/fit |
| DX And | 0 | 70.8% | Count, base information/fit |
| U Gem, hold | 27 | 90.6% | Count, eclipse guard inconclusive |
| IP Peg, hold | 0 | 66.7% | Count, windows, base information/fit, eclipse guard inconclusive |

**Attrition.** In the ordinary report-level diagnostic, 822 clock-available pre-1990 forecasts fell to 280 with a covered previous interval and 274 after requiring a mark. Observing holes, ineligible onset windows and association all contributed to the missing history.

**SS Aur.** Separate parsing found 252 positive visit medians in the flagged run, of which 176 lay in the band $12.5 < m \leq 13.5$, just fainter than the decisive threshold. Its decisive-bright nights form four groups of one, seven, three and five nights. Removing this flag would leave SS Aur the only star above the count gate.

The eligible-window fraction and supported count of every star are shown in Fig. 7.

## Appendix F: Precision planning

Synthetic studies asked how many stars and forecasts a multi-star test would need. They simulated development fits, guarded marks, delayed labels and rolling level updates in Gaussian log gaps, with a nominal per-star gain of 0.004 nats before mark noise and learning. Two statistics answer different questions. For a unit-variance Gaussian model with standardised mark $x$, true coefficient $\beta$ and frozen forecast shift $bx$,

$$E[U_0] = \beta, \qquad E[\Delta s] = b\beta - b^2/2,$$

so a positive score moment $U_0$ can accompany a negative gain $\Delta s$ by the actual forecast; with $\beta = 0.05$ and $b = 0.20$, the expected gain is $-0.010$ nats.

**Table F1.** Whole-star rejection frequencies at a nominal per-star gain of 0.004 nats.

| Stars and condition | Nominal evaluation issuances | Score-moment rejection | Frozen-gain rejection |
|---|---|---|---|
| Eight, clean | 1,100 | 81.8% | 36.7% |
| Eight, persistent / small development / missing marks | 1,100 | 63.6% | — |

| Stars and condition | Nominal evaluation issuances | Score-moment rejection | Frozen-gain rejection |
|---|---|---|---|
| Eight, noisy / delayed / common state | 1,100 | 38.7% | — |
| Six, clean | 1,100 | 76.8% | 33.1% |
| Six, clean | 2,000 | 94.6% | 48.8% |
| Six, persistent / small development / missing marks | 1,100 | 59.4% | 21.3% |
| Six, noisy / delayed / common state | 1,100 | 36.2% | 10.3% |

The grids used 70,000 and 16,000 trials. Neither establishes a robust sample size across its stressed conditions.

## Appendix G: Prior art

CM92's Table 4 gives, for 705 outbursts from 1896 to 1992, correlations of duration with the following onset-to-onset cycle of 0.475 and with the preceding cycle of 0.145; for end-to-end cycles the pattern reverses (0.093 and 0.521). An onset-to-onset cycle contains its first outburst's duration, so this study's target is the quiet interval alone. Durations split at 12 days into 401 short and 304 long outbursts. A bounded independent search found no earlier preregistered probabilistic held-out recurrence evaluation; its queries and receipts are in `data/provenance/a10m_novelty/`.

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401 **Fig. 1.**

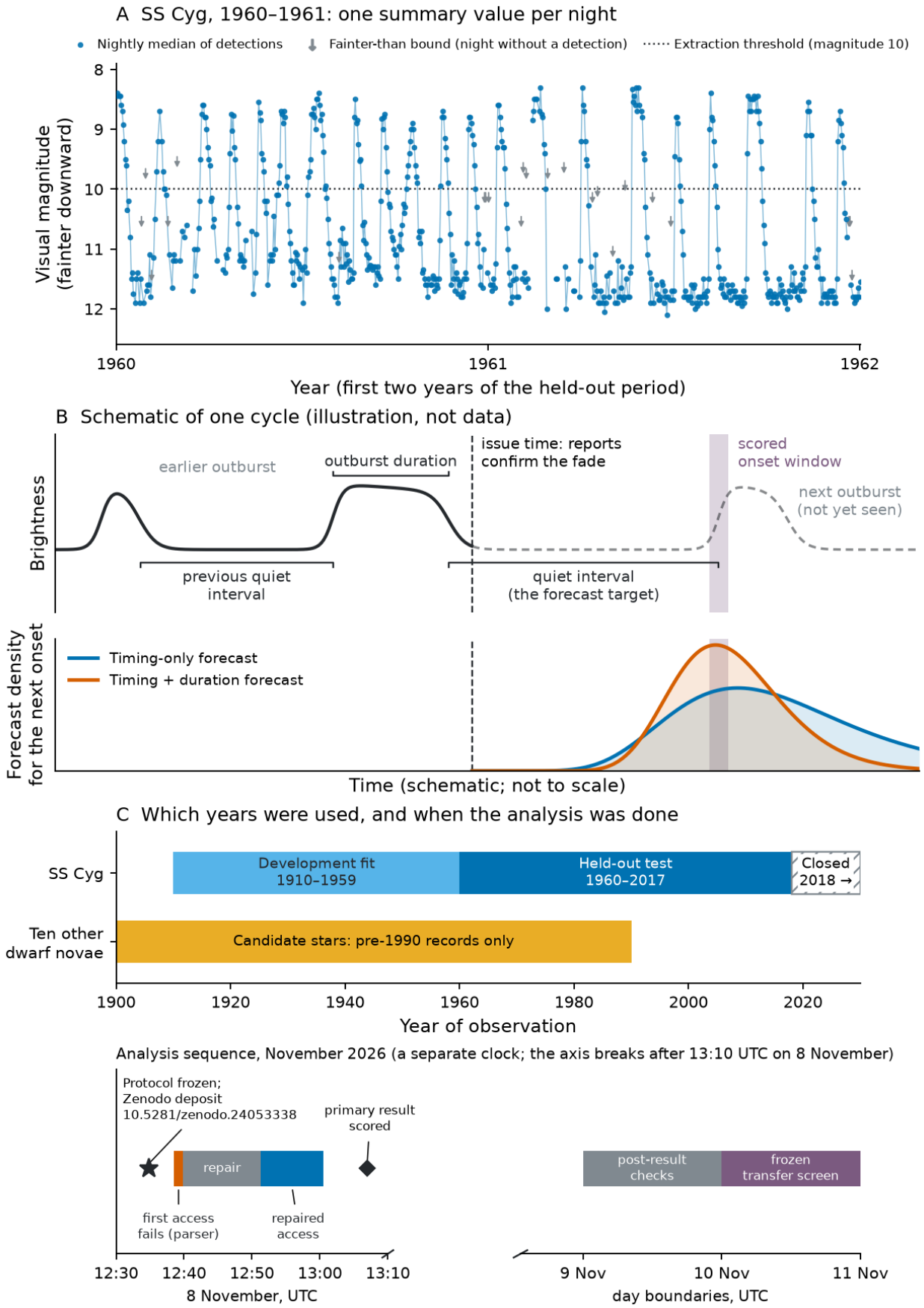



Figure 1: Context. (A) The first two years of the held-out period: one value per night, the median of observers' detections, with fainter-than bounds on nights without a detection; the dotted line is the magnitude-10 extraction threshold. Long and short outbursts alternate between a quiescent level near magnitude 12. (B) Schematic of one cycle. When reports confirm that an outburst has faded, a forecast is issued for the quiet interval that follows. Both models use the previous quiet intervals; the duration model also uses the length of the outburst just completed. The working score is the log probability each forecast assigns to the extracted onset window of the next outburst. (C) Upper: the observation years used for development, testing and the transfer screen. Lower: the November 2026 analysis sequence, on a separate clock: protocol deposit, first

**Fig. 2.**

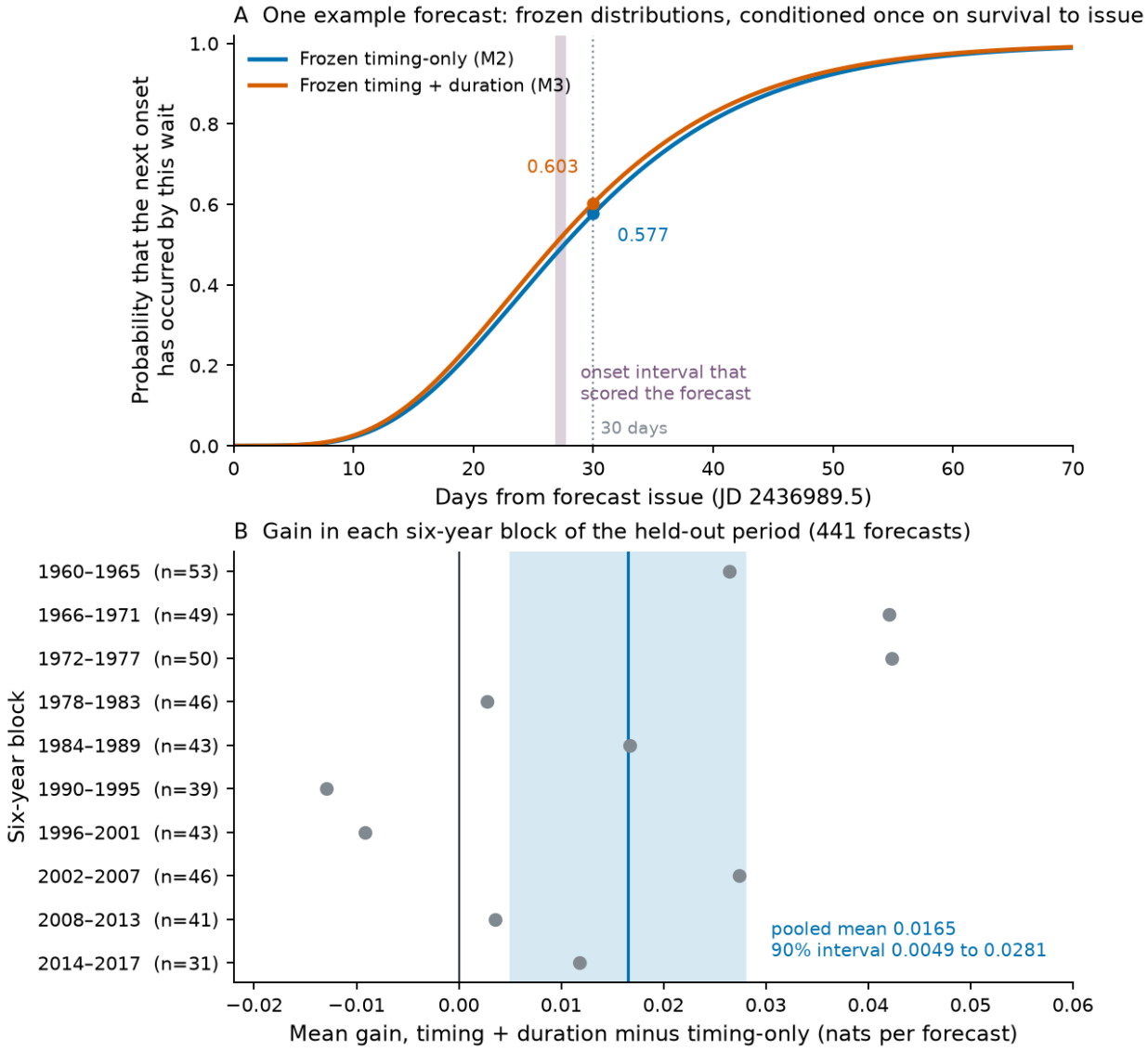



Figure 2: The primary result. (A) The first eligible forecast of the held-out period, chosen by date: cumulative probability of the next onset for the frozen timing (M2) and timing-plus-duration (M3) models, conditioned on survival to issue, with the extracted onset window that scored it. M3 placed 0.603 of its probability within 30 days against 0.577 for M2. (B) Mean paired gain in each six-year block, with the pooled gain of 0.0165 nats per forecast and its 90% interval [0.0049, 0.0281] over 441 forecasts. Block means are descriptive.

403 **Fig. 3.**

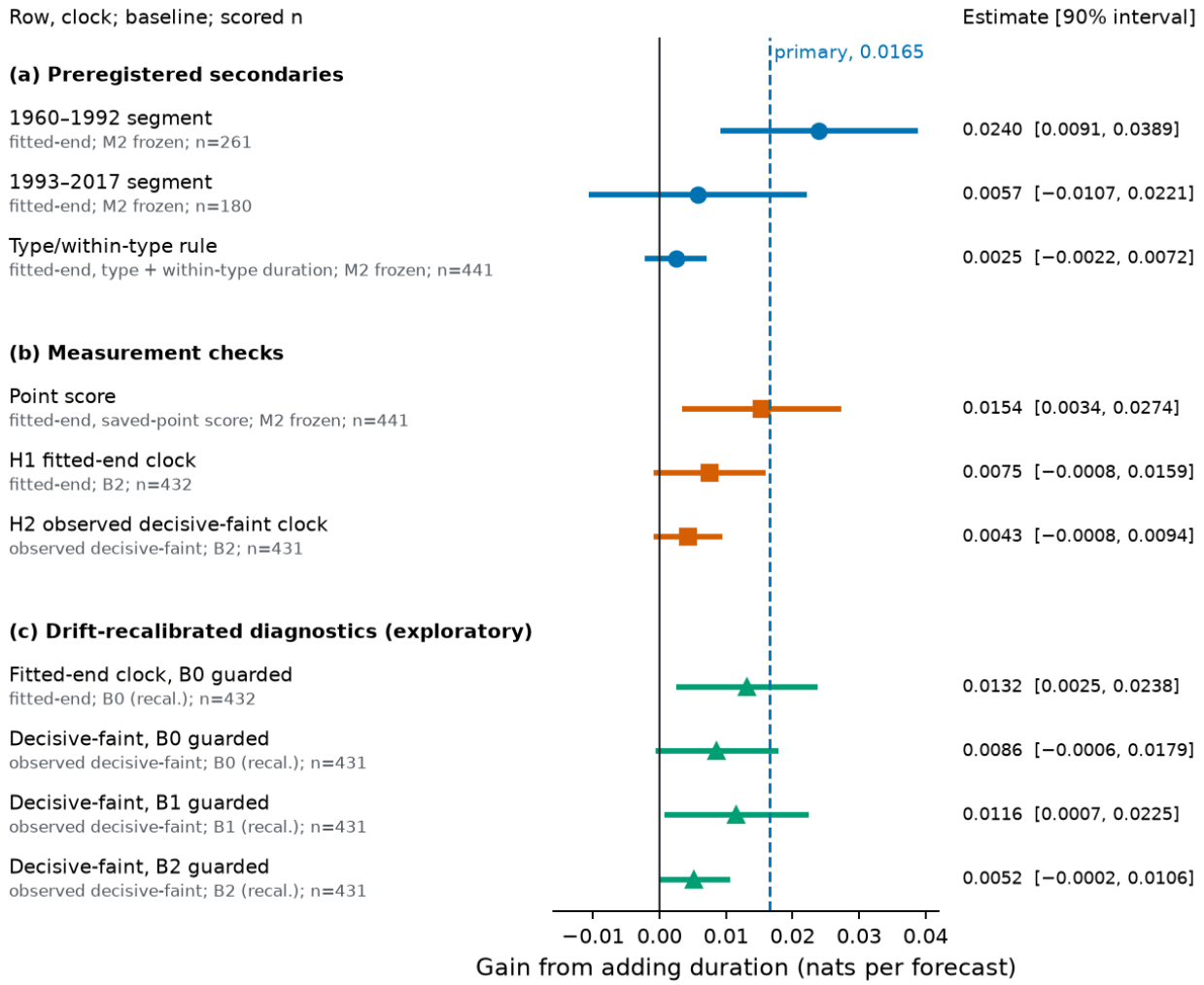



Figure 3: Duration gains from every comparison, with the primary gain as the dashed reference. Each row gives its clock, baseline and number of scored forecasts. (a) The preregistered era split and long/short rule, frozen M2 baseline. (b) Rescoring against point onsets, and the report-window comparisons on the fitted-end clock (H1) and the decisive-faint clock (H2), both with the state-space baseline B2. (c) Drift-recalibrated ("guarded") diagnostics, with duration recentred on the preceding ten outbursts and the baseline level updated from preceding outcomes; B0 uses recent-interval covariates and B1 discounted history. Intervals are 90%, from six-year clusters. Groups (b) and (c) reuse the opened data.

**Fig. 4.**

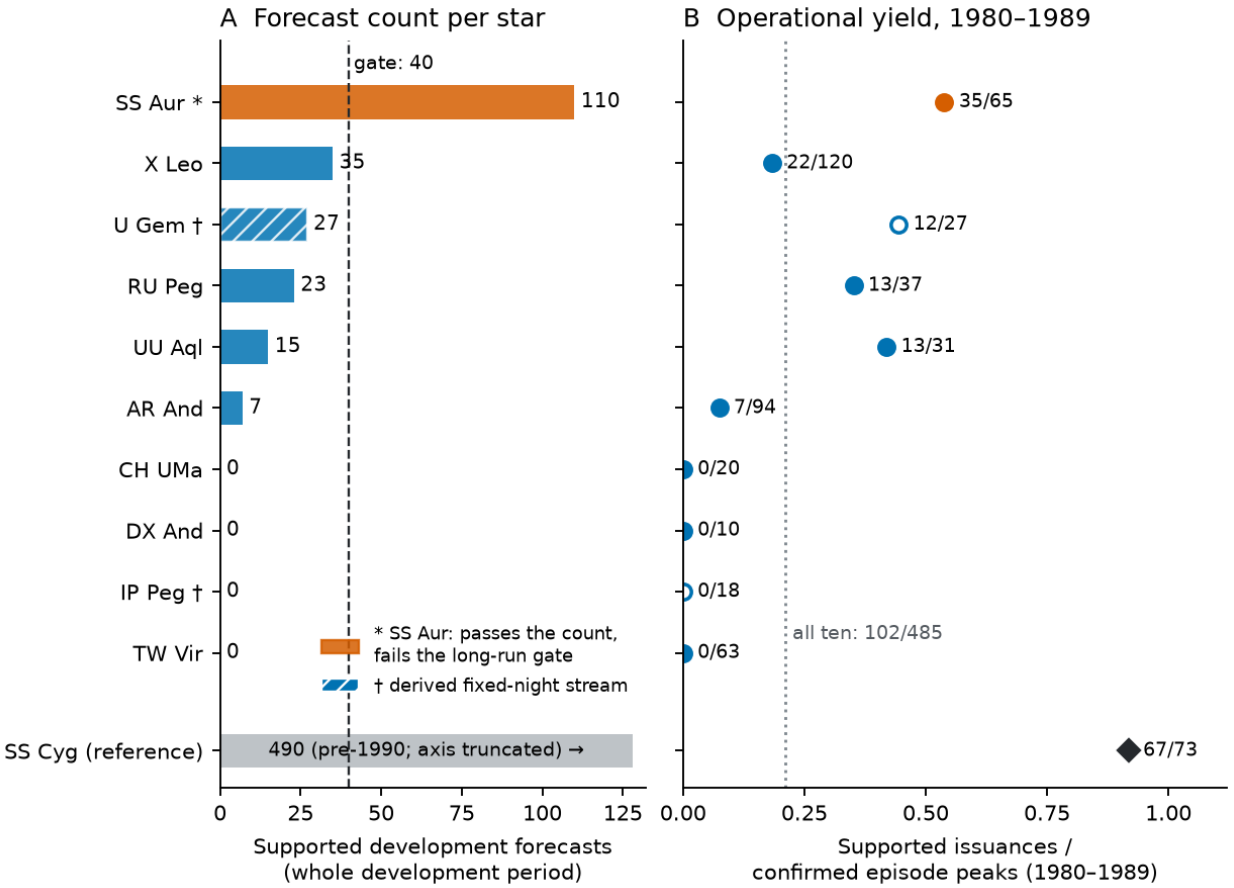



Figure 4: The frozen transfer screen. (A) Supported forecasts per candidate star in its pre-1990 record, against the gate of 40. SS Aur passes the count and fails the long-run check. U Gem and IP Peg use derived fixed-night streams. SS Cyg's pre-1990 count under the same chain is shown for reference, with the axis truncated. (B) Supported issuances divided by confirmed episode peaks, 1980–1989, with counts beside each point; the dotted line is the ten candidates pooled (102/485) and the diamond SS Cyg (67/73). Numerators count supported forecasts by issue date; denominators count confirmed episodes by peak date within the same decade. Hold stars use the specified derived-night streams. Ratios describe operational yield under these rules.

405 **Fig. 5.**

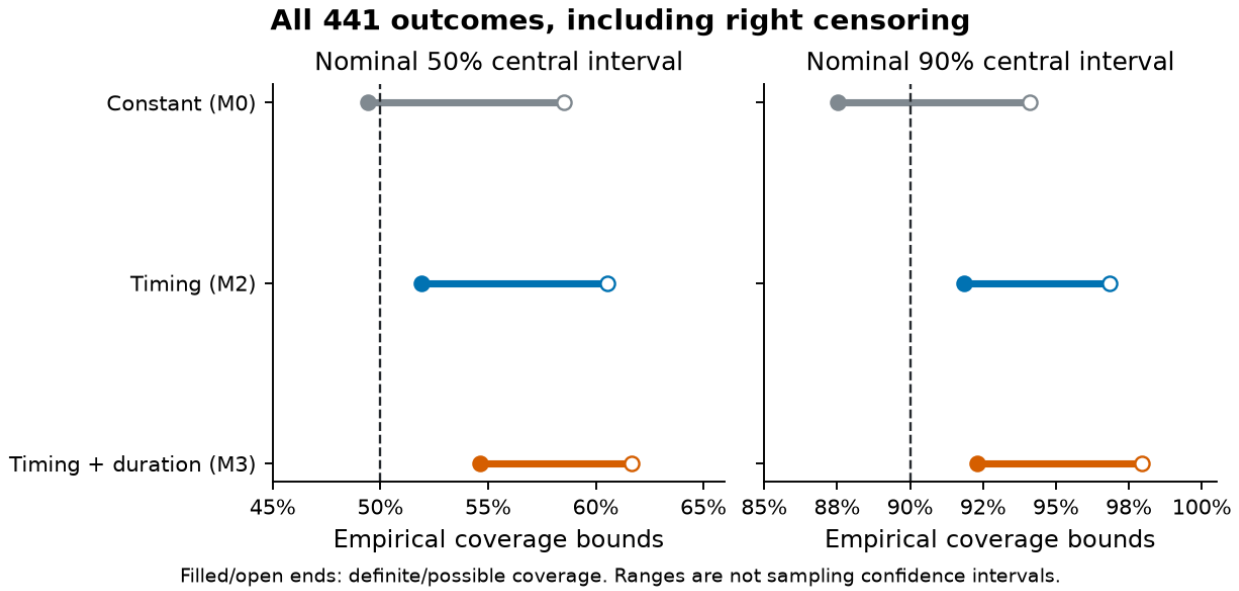



Figure 5: Coverage of central predictive intervals for the constant (M0), timing (M2) and timing-plus-duration (M3) forecasts over all scored forecasts, 1960–2017. Each segment runs from definite to possible coverage given the extracted onset windows and censoring; dashed lines mark the nominal fractions.

406 **Fig. 6.**

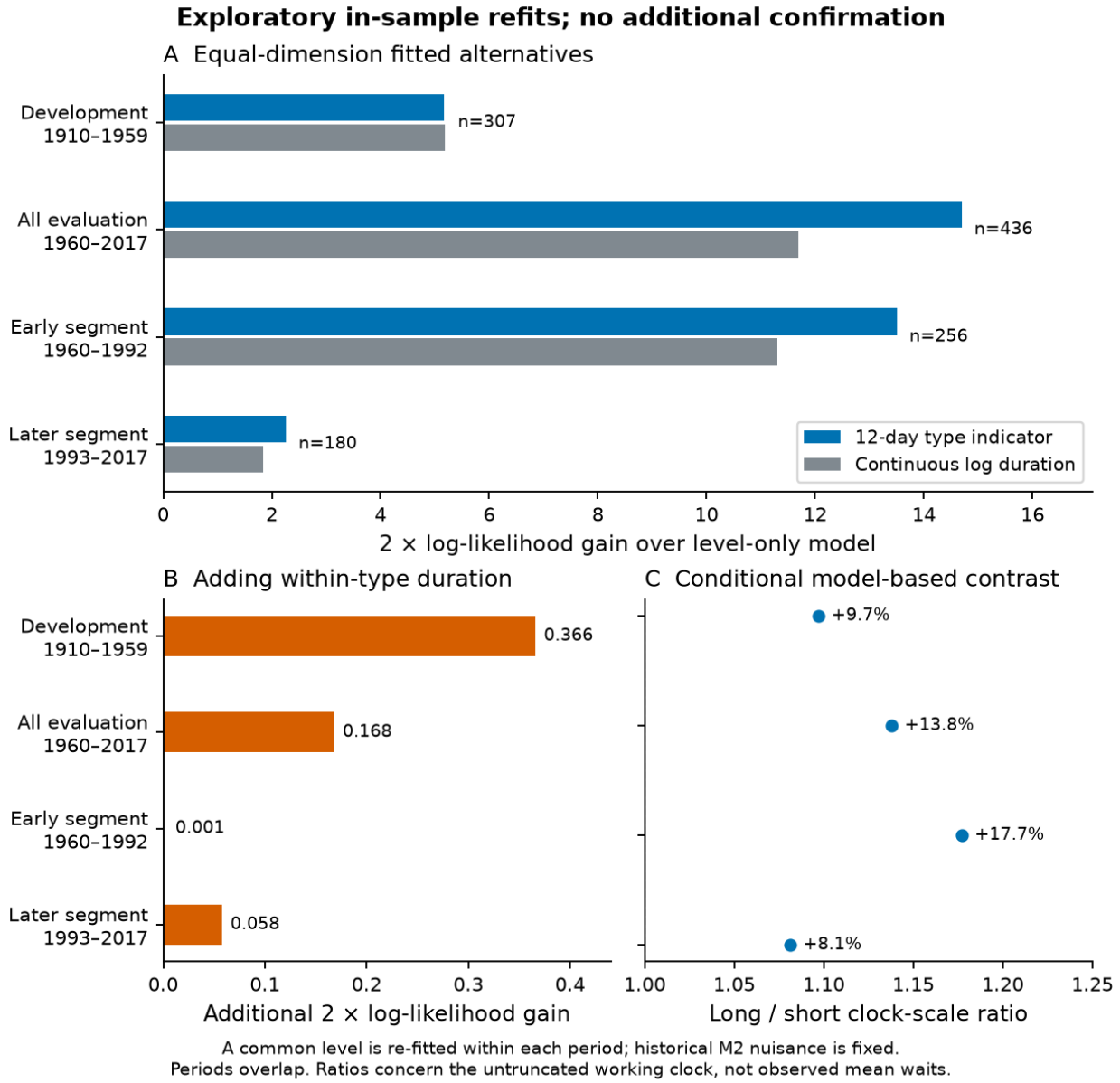



Figure 6: Exploratory long/short comparisons fitted to the evaluation outcomes, with M2's historical parameters fixed and a free common level. Left: likelihood-ratio statistics for the class indicator (14.7), continuous log duration (11.7) and within-class duration added to class (0.17). Right: long/short ratio of the untruncated working-clock scale by period.

**Fig. 7.**

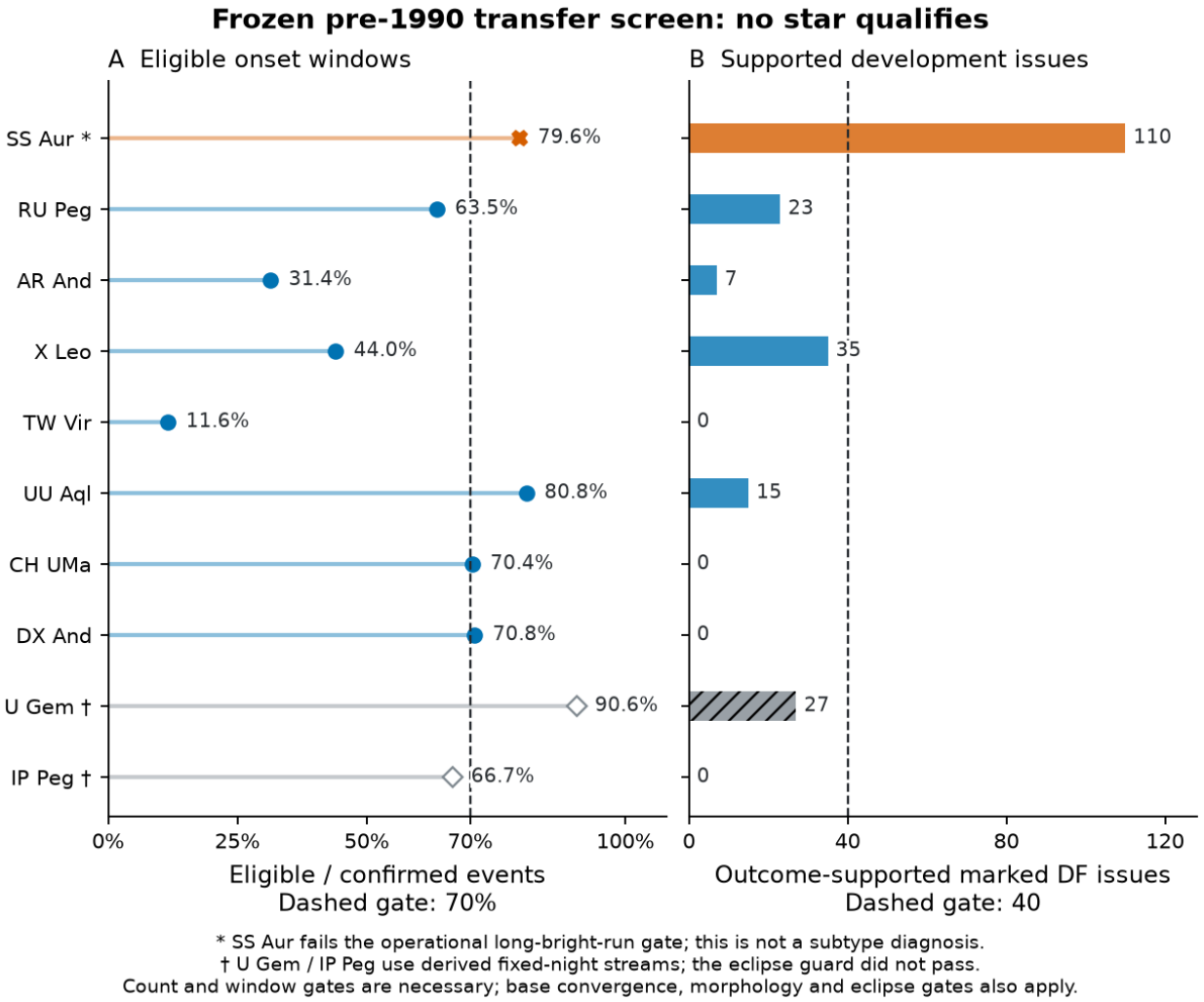



Figure 7: Frozen pre-1990 transfer screen: eligible onset-window fraction against supported development forecasts for the ten candidates, with the frozen thresholds of 70% and 40. SS Aur passes both and fails the long-run check; U Gem and IP Peg use derived night streams.